

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/02

Paper 2 Structured Questions

13 September 2018

Candidates answer on the Question Paper.

No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Total	

This document consists of **24** printed pages.

[Turn over

Answer **all** the questions in this section.

- 1 (a) A clinic sequenced the β -globin gene locus of five different patients and tabulated the results in the following table.

Table 1. Genetic profile of the 6th codon of patients' β -globin gene

Patient	DNA codon sequence*	Change in amino acid
1	GAG	Glu (unchanged)
2	GAA	Glu (unchanged)
3	GTG	Glu $\rightarrow$ Val
4	GAC	Glu $\rightarrow$ Asp
5	GTG	Glu $\rightarrow$ Val

* DNA sequence on the 6th codon of the human β -globin gene

Based on the information in **Table 1**,

- (i) Which patient(s) has/have the sickle-cell anaemia mutation?

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[1]

- (ii) Explain how the change of amino acid would result in a sickle shaped red blood cell.

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[3]

- (iii) Suggest and explain why patient 4 does not have sickle cell anaemia.

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[2]

Fig. 1 shows a linear chromosome undergoing the first round of DNA replication. The arrows show the direction of synthesis for the daughter DNA strands. Ends A, B C and D represent the ends of the newly synthesized daughter DNA strands.

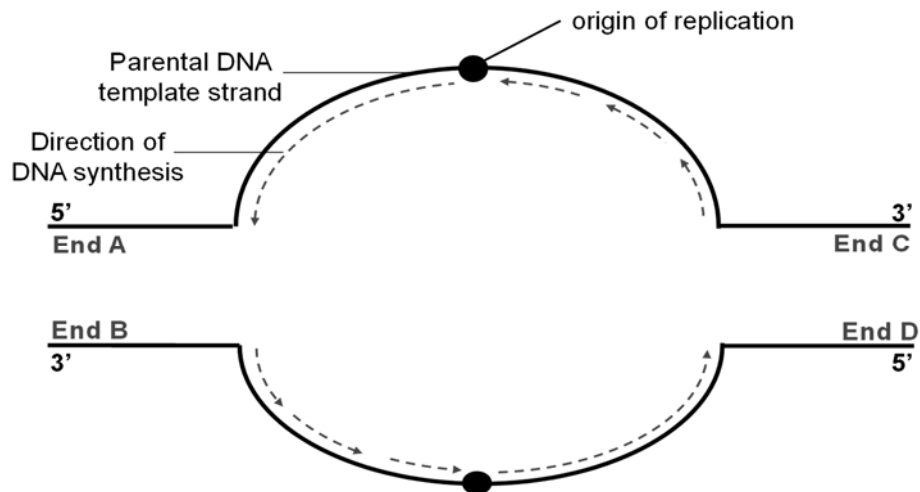


Fig. 1

(b) With reference to **Fig. 1**,

- (i)** state which end(s) will experience the end-replication problem after the first round of DNA replication has been completed.

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 [1]

- (ii)** suggest one modification that a researcher could do to the parental DNA strands in order to completely remove the end-replication problem.

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 [1]

- (c)** Telomerase is an enzyme that is found in certain cell types and it helps to ameliorate the end replication problem by lengthening the ends of the chromosomes, which are also known as telomeres.

Explain why the telomerase enzyme does not prevent the end-replication problem from occurring despite being able to lengthen the telomeres.

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 [2]

[Total: 10]

2 Fig. 2.1 shows some *Allium* sp. plant cells in various stages of the mitotic cell cycle.

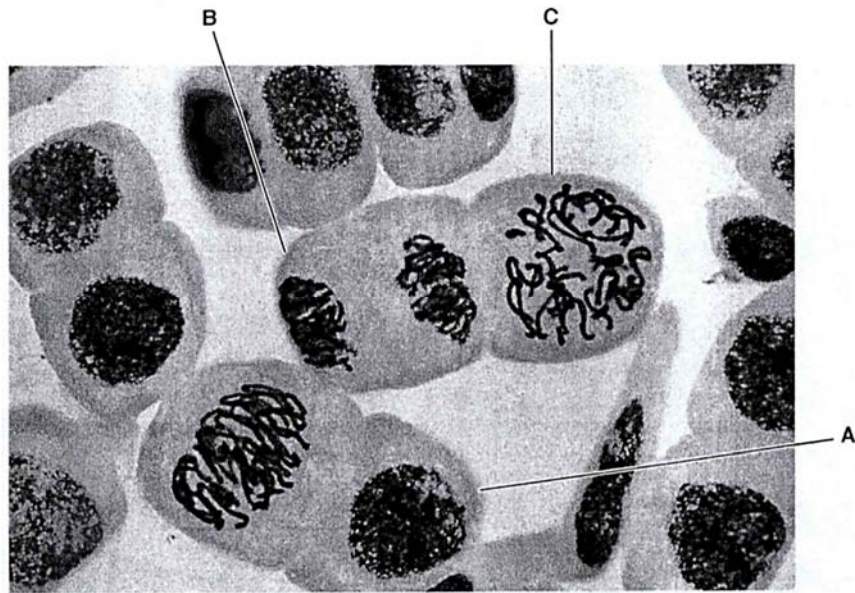


Fig. 2.1

(a) (i) Identify the three stages shown by the labelled cells.

A

B

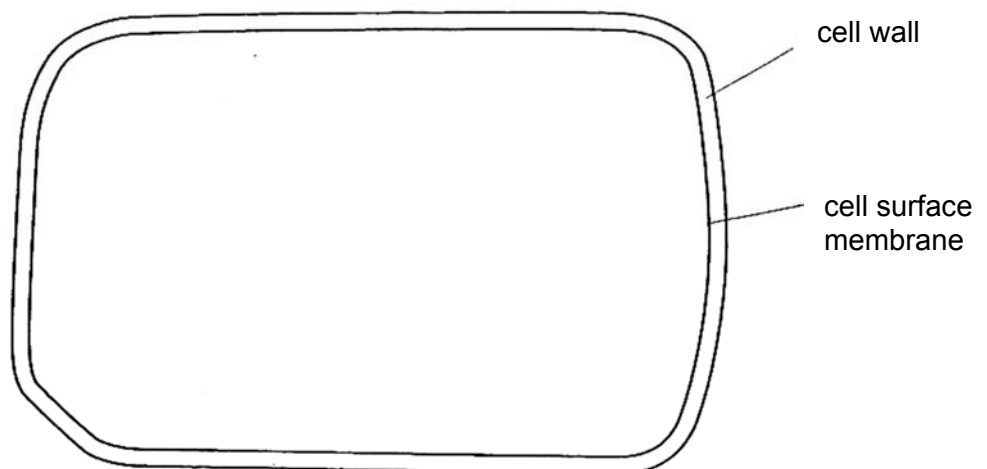
C

[3]

(ii) Identify the stage of mitosis that follows that shown in cell C.

[1]

(iii) In the cell outline below, draw and label the structures visible in a cell that is in the stage you have named in (ii). $2n$ for this plant is 6.



[3]

- (b) Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This is a reversible competitive inhibitors of one of the enzymes in the metabolic pathway that results in the formation of purines.

Explain how the use of Methotrexate will slow down the mitotic cell cycle.

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[2]

- (c) Prokaryotic organisms such as *Escherichia coli* divide by simple cell splitting (binary fission), not mitosis.

Apart from ribosomes, prokaryotes have no organelles comparable to those found in eukaryotes and have a circular 'chromosome' with no centromere.

With reference to the information above and your knowledge of mitosis, suggest why mitosis does **not** occur in prokaryotes.

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[2]

[Total: 11]

- 3 The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced by the action of the enzyme tyrosinase in cells called melanocytes. Low enzyme activity leads to the production of red pigment, while high enzyme activity brings about black pigment production.

The activity of the enzyme is increased when melanocyte stimulating hormone (MSH) combines with a MSH receptor. The receptor is coded for by the gene, **R**, which has three alleles, **R^D**, **R^A** and **r**. **R^D** and **R^A** each codes for a receptor with a different activity. No receptor is produced by the recessive allele, **r**.

The dominant allele of a second gene, **B**, codes for a protein which binds to and blocks the MSH receptors coded for by **R^A**, thus preventing stimulation of tyrosinase activity in a melanocyte. The receptors coded for by **R^D** is insensitive to the protein coded by **B**. The recessive allele, **b**, does not produce a functional protein.

- (a) (i) State the name given to the interaction between the **R** and **B** gene loci.

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..... [1]

- (ii) Explain why animals with the genotype **R^AR^ABB** have red coats.

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..... [1]

- (iii) A red cow, with genotype $R^A R^A B B$ is mated with a bull which is homozygous recessive at both gene loci.

Draw a genetic diagram in the space below to show the expected genotypes and phenotypes and their ratios in the F_1 and F_2 generations.

[5]

During a health screening exercise of cattle in a farm, the height of the bulls was measured and the data collected is shown in **Table 3**.

Table 3

Height/cm	Number of bulls
131—135	3
136—140	9
141—145	21
146—150	12
151—155	2

- (b)** Distinguish between the two types of variation shown in coat colour and height in the Norwegian cattle.

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[3]

In the honey bee colony, the queen bee is solely responsible for laying eggs and the drones for fertilizing her. The worker bees have well-developed mouthparts and structural adaptations for collecting nectar and pollen to gather food and to perform other duties in the hive. Male bees are developed from haploid eggs while both queen and worker bees develop from fertilized eggs.

- (c)** Explain how the phenotypic differences between the queen and the worker bees come about despite both being developed from fertilized eggs.

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[2]

[Total: 12]

4 Fig. 4.1 shows a cyclic series of reactions that occurs during photosynthesis.

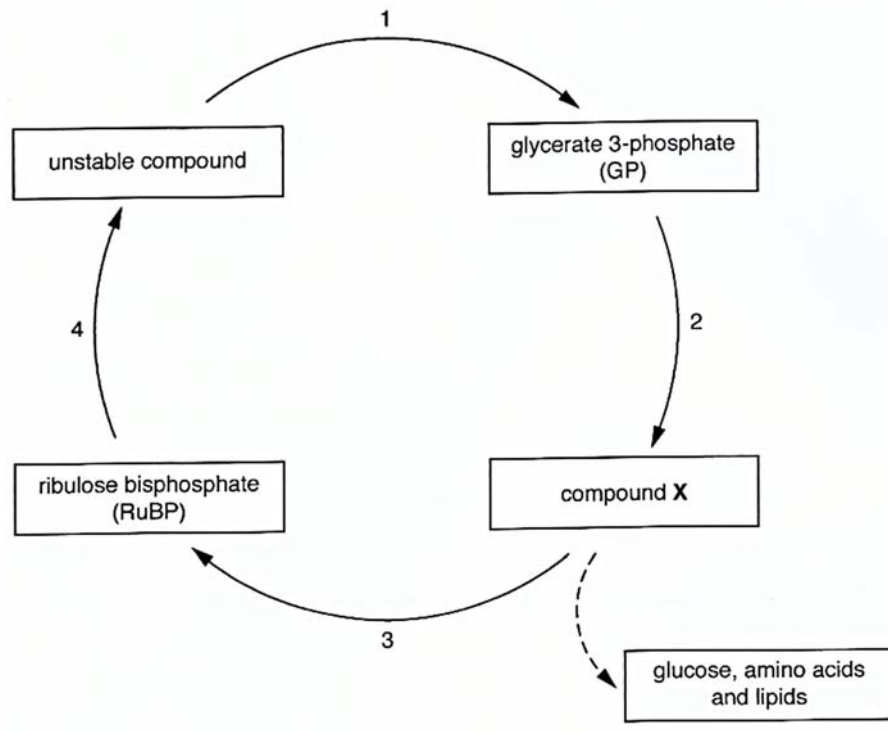


Fig. 4.1

(a) (i) State the name of this cycle.

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 [1]

(ii) State the number of carbon atoms in one molecule of compound X.

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 [1]

(iii) Give the label number of each solid arrow that represents reactions where energy is released from ATP.

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 [1]

(iv) Give the label number of the solid arrow that represents reactions where carbon dioxide is taken up.

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 [1]

(b) Fig. 4.2 is an electromicrograph of a chloroplast.

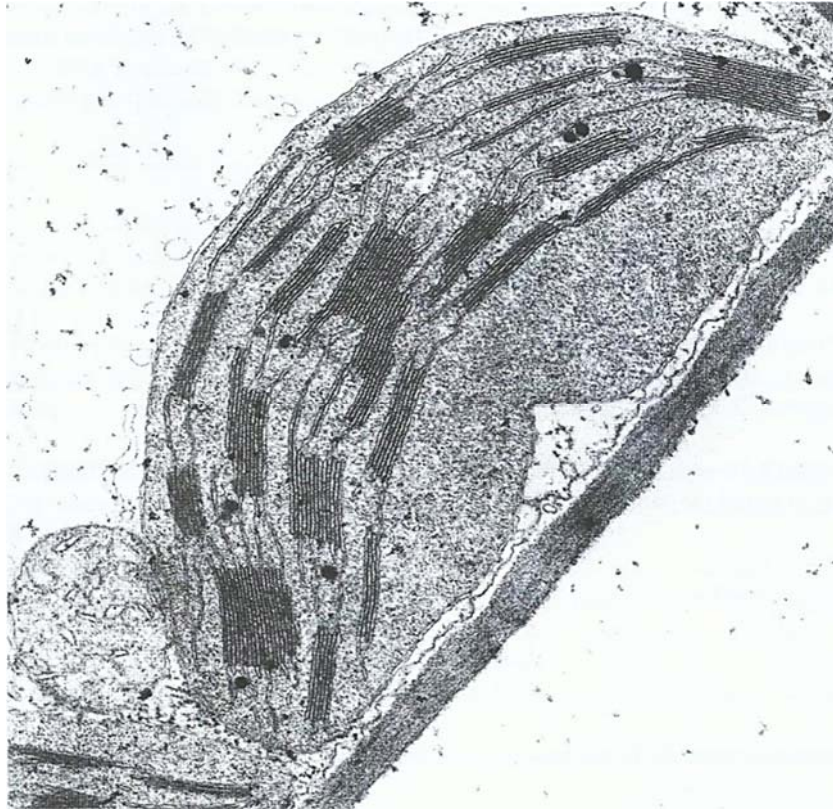


Fig. 4.2

- (i) On Fig. 4.2 use the letter Y and a label line to identify where the cyclic series of reactions shown in Fig. 4.2 occurs.

[1]

- (ii) Chloroplasts and mitochondria both have double membranes.
State two other structural similarities between chloroplasts and mitochondria.

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[2]

(c) The enzyme rubisco has an important role in the cycle of reactions shown in **Fig. 4.1**.

Both oxygen and carbon dioxide can bind to the active site of rubisco.

- (i) With reference to the information above, suggest why an increase in carbon dioxide concentration increases the rate of reactions shown in **Fig. 4.1**.

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[2]

- (ii) At low temperatures, the activity of rubisco is low and this limits the rate of photosynthesis. Some plants that have a high rate of photosynthesis at low temperatures have an enzyme, **Z**, which increases the activity of rubisco.

Suggest how scientists could make use of this information to improve a crop plant so that it could be grown in cold conditions.

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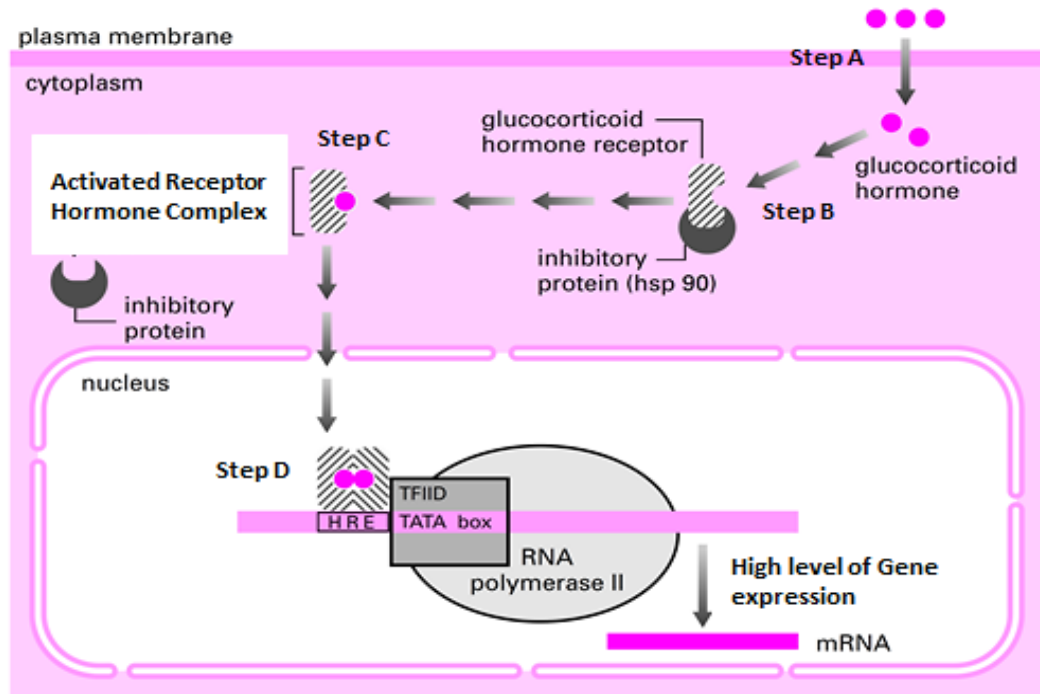
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[2]

[Total: 11]

- 5 Glucocorticoids are steroid hormones produced by the adrenal cortex that increase the transcription of several genes important in carbohydrate and protein metabolism. **Fig. 5.1** below shows how glucocorticoids can pass through the plasma membrane to enter the cytosol and bind to glucocorticoid hormone receptors. In the absence of glucocorticoid hormone, its receptor remains in the cytosol and is inactive.



Hormone response element (HRE)

Fig. 5.1

- (a) (i) Glucocorticoid hormone receptor is a class of transcription factors. With reference to **Fig. 5.1**, explain how receptor hormone complex (Step C) can be activated.

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[2]

- (ii) Suggest how glucocorticoid hormone receptor is able to bind to the specific region of the DNA known as HRE site shown in step D of **Fig. 5.1**.

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[2]

- (iii) With reference to **Fig. 5.1**, explain how high level of gene expression may be regulated by activated receptor hormone complex within the nucleus.

[3]

Fig. 5.2 below shows the various steps involved in the processing of primary RNA transcript.

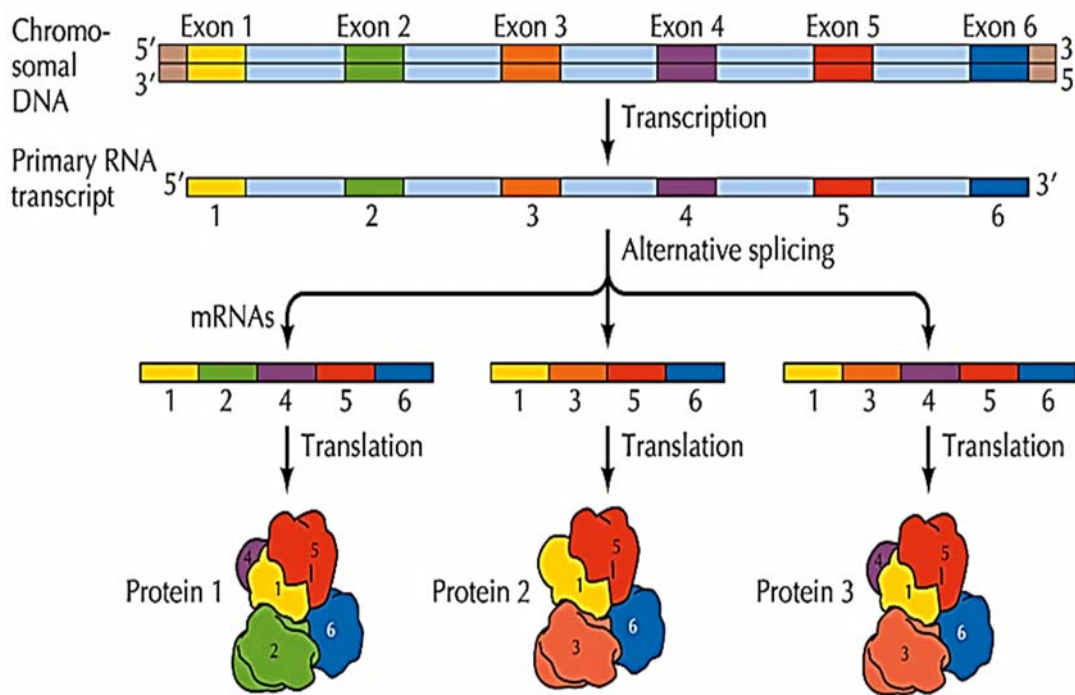


Fig. 5.2

- (b) (i) Name process A and explain why it is essential in eukaryotic cells from an evolutionary view point.

[3]

- (ii) Prior to completion of step A, the primary RNA is modified in several important ways. Explain how **one** of these ways helps to regulate gene expression.

[1]

- (iii) With reference to **Fig. 5.3**, label the following region (label only once):

- (i) 5' UTR
- (ii) 3' UTR
- (iii) coding sequence
- (iv) promoter

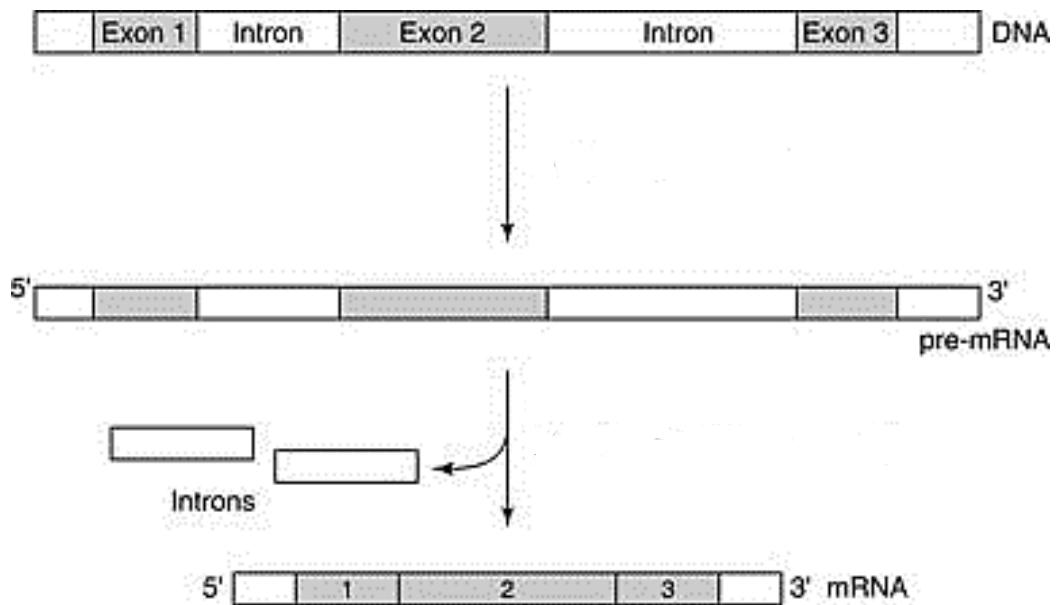


Fig. 5.3

[1]

[Total: 12]

6 Viruses employ various mechanisms to evade the host immune response.

The presence of Major Histocompatibility Complex (MHC) class I and class II molecules on Human Immunodeficiency Virus (HIV) has been shown to render the virions more infectious.

(a) Suggest how HIV acquires the MHC molecules.

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[1]

Fig. 6.1 shows the number of T helper cells in the blood and the number of HIV viruses in the body over the course of an untreated HIV infection.

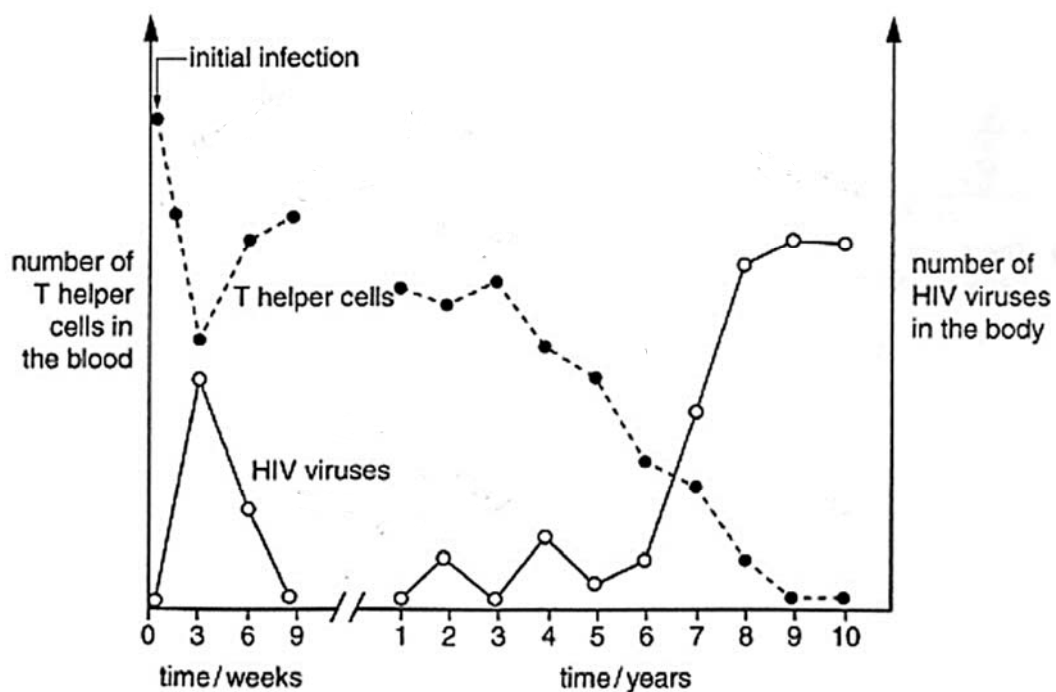


Fig. 6.1

(b) (i) Describe the changes shown in **Fig. 6.1**.

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[3]

(ii) Suggest how the changes in the number of T helper cells shown in **Fig. 6.1** would affect the health of an untreated HIV-infected individual over the course of the infection.

[3]

(c) Although a person may be infected only once by one strain of HIV he may eventually have many strains of HIV. Briefly explain why.

[3]

[Total: 10]

- 7 (a) In bacteria, it is common for two or more genes to be arranged together in an operon.

(i) Briefly discuss the arrangement of genetic sequences within an operon.

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[2]

(ii) State the biological advantage of an operon organization.

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[1]

- (b) The sequential use of two sugars by a bacterium is known as diauxic growth. It is a common phenomenon among many bacterial species. When glucose is one of the two sugars available, it is typical that the bacterium metabolises glucose first, and then a second sugar after the glucose has been used up. Among *E. coli* and related species, diauxic growth is regulated by intracellular cAMP levels and the catabolite activator protein (CAP).

(i) Summarise the effects of glucose and lactose on the ability of the lac repressor and the cAMP-CAP complex to regulate the lac operon.

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[3]

You have isolated a mutant strain of *E. coli* in which the lac operon is constitutively expressed. To understand the nature of this defect, you create a merozygote (diploid cell) in which the mutant strain contains an F' factor with a normal lac operon and a normal lacI gene. You then compare the mutant strain and the merozygote with regard to their β -galactosidase activities in the presence and absence of lactose.

You obtain the following results:

	Addition of lactose	Amount of B-galactosidase (percentage of mutant strain in the presence of lactose)
Mutant	No	100
Mutant	Yes	100
Merozygote	No	100
Merozygote	Yes	200

Fig. 7

(ii) Explain the nature and effect of the defect in the mutant strain.

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[2]

[Total: 8]

- 8 *Hebe* species are native shrubs found throughout New Zealand, particularly on the banks of lowland streams and rivers, as well as in the alpine (above the treelines) and subalpine zones of the mountains.

Table 8.1 shows the description of seven *Hebe* species.

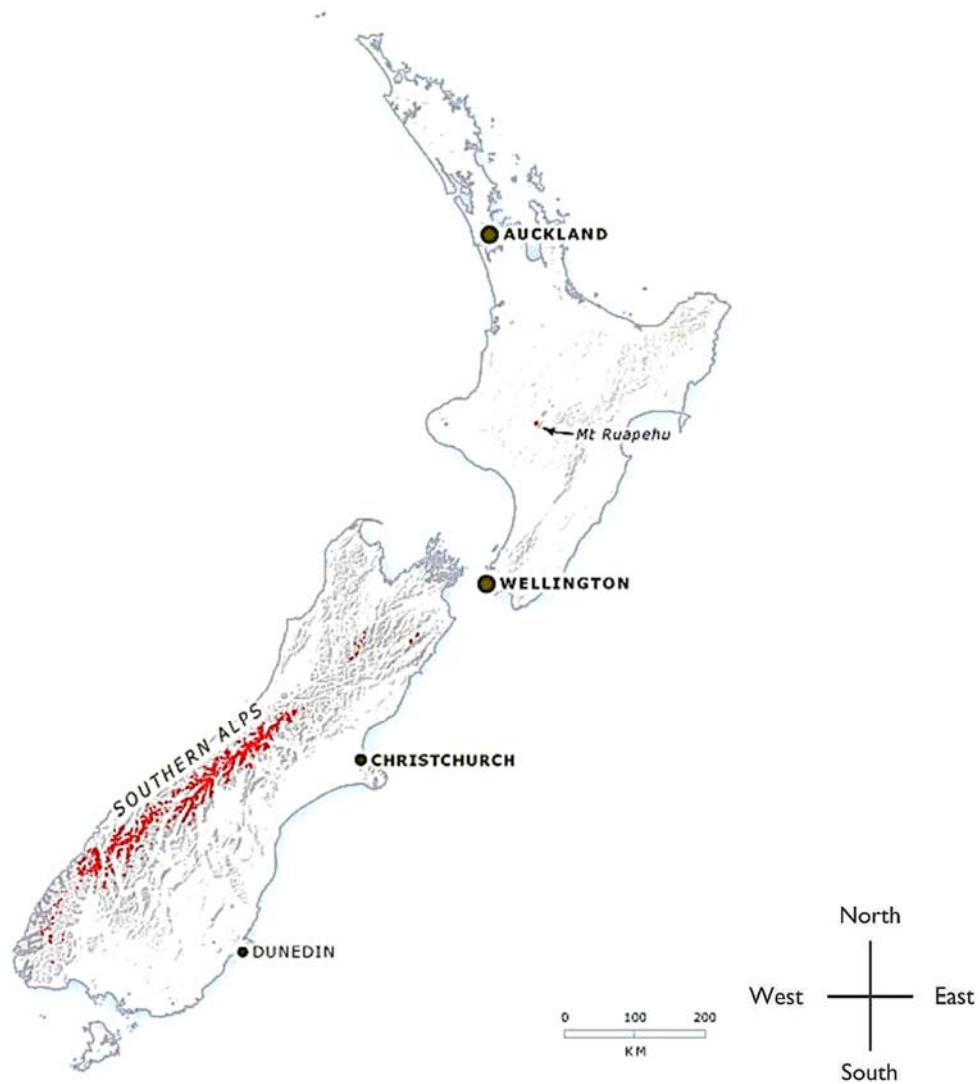
Table 8.1

Species	Morphology	Habitat	Number of chromosomes in a gamete
<i>H. imbricata</i>	Rounded shrub up to 0.6 m tall.	Drier mountains of the South Island.	20
<i>H. raoulii</i>	Small low growing shrub up to 0.3 m tall.	Drier mountains and hills of the South Island.	21
<i>H. crenulata</i>	Low shrub up to 1 m tall.	Subalpine shrubland and grassland, often in shallow or rocky soils of northern South Island.	40
<i>H. elliptica</i>	Bushy shrub up to 2 m tall.	Seaside, west coast of South Island.	20
<i>H. cupressoides</i>	Shrub up to 2 m tall.	River-flats and terraces, as well as subalpine east of the South Island.	21
<i>H. gracillima</i>	Shrub up to 2 m tall.	Damp swampy places in the west coast of South Island.	40
<i>H. topiaria</i>	Neatly rounded bushy shrub up to 2 m tall.	Wet subalpine areas in the north of South Island.	61

It has been proposed that the ancient population that gave rise to the *Hebe* species in New Zealand most likely came from *H. cupressoides* in Australia about five million years ago.

The movement between the Australian Plate and Pacific Plate over the last five million years resulted in the formation of the Southern Alps. The rise of the Southern Alps caused the west coast of the South Island to be much wetter than the east coast. The map of New Zealand is shown in **Fig. 8.1**.

Between 1.6 million and 10,000 years ago, New Zealand was exposed to cold glacial and warmer interglacial periods. These caused the sea levels to fluctuate by up to 135 m. In the glacial periods, forests retreated from the mountains towards the coasts. This also changed the soil quality in the regions.

**Fig. 8.1**

(a) Explain why the population is the smallest unit that can evolve.

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[3]

- (b) (i) Explain how the environment factors contributed to the formation of different species of *Hebe*.

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[3]

- (ii) Suggest how the 61 chromosomes of *H. topiaria* could have arisen.

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[3]

- (c) Suggest **two** reasons why fossils of whole plant are difficult to obtain, as compared to animal fossils.

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[2]

[Total: 11]

- 9 The toxins released by some pathogenic bacteria can be altered chemically so that they are harmless. These harmless toxins are called toxoids.

Toxoids are used in vaccines to provide protection against some infectious diseases.

Describe the response of the immune system to the injection of a toxoid.

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[5]

[Total: 5]

- 10** Reef building corals are marine invertebrates found in shallow, clear tropical seas. The corals secrete an exoskeleton of calcium carbonate that becomes the underlying structure of the coral reef.

Zooxanthellae are a group of unicellular photosynthetic algae that live inside the cells of reef building corals. The relationship is beneficial to both the zooxanthellae and the coral.

- (a)** Evidence shows that the relationship between zooxanthellae and the reef building coral has evolved by free living algae invading corals that did not contain algae.
- (i)** Corals that do not need zooxanthellae can live at greater depth than reef building corals.

Explain why.

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[3]

- (ii)** Suggest how the zooxanthellae may benefit in two ways from their association with the corals.

Benefit 1

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Benefit 2

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[2]

Under conditions of environmental stress the relationship between the reef building corals and the zooxanthellae can break down. Loss of zooxanthellae and the subsequent whitening that occurs is known as coral bleaching. Coral bleaching can lead to death of the coral.

- (b)** State **one** reason why permanent loss of zooxanthellae can lead to death of the coral.

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[1]

- (c) One type of environmental stress that can cause coral bleaching is an increase in sea temperature.
- (i) Suggest why areas of sea with reef building corals are particularly susceptible to increased temperature as a result of global climate change.

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[2]

- (ii) The optimum temperature range for the survival of reef building corals is 25°C to 29°C.

Explain why reef building corals will be affected by an increase in temperature above the optimum.

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[2]

[Total: 10]